

# Biodegradable and Non-biodegradable waste classification for Smart Waste Management

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**Abstract**— Effective waste management is a growing concern in urban and rural areas alike, where improper segregation especially of wet waste leads to environmental degradation, unpleasant odors, health hazards and increased municipal workload. Manual segregation is labor-intensive, inconsistent, and unhygienic, prompting the need for an automated solution. This research proposes a smart classification system of wet waste with the help of Deep Learning, specifically Convolutional Neural Networks (CNNs), to distinguish between biodegradable and non-biodegradable waste items. A custom dataset comprising images of various wet waste types such as food scraps, peels, tissues, and other organic/inorganic substances was collected, cleaned, and labeled for supervised learning. Image pre-processing techniques like resizing of images, normalization, and augmentation were done to enhance model robustness. The CNN model was designed and trained to extract visual features and classify the input waste images with high accuracy. The proposed model is suitable for resource constrained environments like IoT-enabled smart bins, waste sorting stations, or municipal segregation systems. The research bridges a critical gap in waste classification by addressing the underrepresented category of wet waste and contributing to the larger goal of sustainable smart cities.

**Keywords**—Municipal Solid Waste (MSW), Waste Classification, Convolutional Neural Networks, Sustainability

## I. INTRODUCTION

As per annual report on Implementation of Solid Waste Management Rules, 2016 by Central Pollution Control Board (CPCB)<sup>1</sup>, the total quantity of Solid waste generated in India is 160038.9 TPD i.e. Tonnes Per Day. This is a humongous quantity and thus, effective waste management is paramount for maintaining a healthy environment, fostering sustainable development, and ensuring the well-being of present and future generations. Waste management comprises a range of practices aimed at reducing, reusing, recycling, and responsibly disposing of waste materials. This critical process not only mitigates environmental pollution and conserves natural resources but also safeguards public health, promotes economic growth, and enhances the overall quality of life. By addressing the challenges of waste generation and

disposal through comprehensive waste management strategies, communities can mitigate environmental degradation, minimize risks to human health, and pave the way for a more sustainable future.

Waste classification plays a pivotal role in efficient waste management systems by providing a structured approach to handling different types of waste materials. By categorizing waste based on its composition, toxicity, and potential for recycling or reuse, authorities can implement tailored disposal methods that minimize environmental impact and maximize resource recovery. Proper classification allows for the identification of hazardous materials, ensuring they are handled and disposed of safely to prevent harm to human health and the environment. Moreover, waste classification facilitates recycling efforts by separating recyclable materials from non-recyclables, promoting circular economy principles and reducing reliance on finite resources. Ultimately, effective waste classification streamlines waste management processes, enhances environmental sustainability, and contributes to a cleaner, healthier planet.

One of the biggest challenges many urban areas face today is the poor segregation of waste especially wet waste, which often includes food leftovers, vegetable peels, and other organic material. Unfortunately, these biodegradable items are frequently mixed with plastics and synthetic packaging, which not only contaminates the waste stream but also makes composting and recycling a lot harder. In most cities, sorting this waste is still done by hand. It's a slow and error-prone process, and more importantly, it exposes workers to unsafe and unhygienic conditions. As the demand for more sustainable urban living grows, so does the need for a better, more reliable system to handle this task.

In this work, we aim at tackling problem of biodegradable and non-biodegradable waste classification which is commonly known as wet and dry waste classification. The core idea is to train image-based models, specifically Convolutional Neural Networks (CNNs), to recognize and

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<sup>1</sup>[https://cpcb.nic.in/uploads/MSW/MSW\\_AnnualReport\\_2020-21.pdf](https://cpcb.nic.in/uploads/MSW/MSW_AnnualReport_2020-21.pdf)

sort waste into two simple categories - biodegradable and non-biodegradable. Such models have potential to be part of a smart waste segregation system which can classify and segregate waste automatically, without human intervention, saving health hazards for the human labor.

This work has following novel contributions:

- A wet waste dataset collected manually
- Thorough analysis of CNN models for the waste classification task

## II. RELATED WORK

Many review articles are available for AI in waste management such as [1], [2], [3] and [4]. [1] reviewed 6 complete waste management systems from year 2014 to 2018. The main findings are that systems cannot handle mixed waste and not focused on determining e-waste. [2] have performed systematic literature review of 85 research studies, published between 2004 and 2019. They have analyzed the application of AI in various Solid Waste Management (SWM) fields such as level detection for waste bins, forecasting of waste characteristics, parameters prediction for waste process, routing of collection vehicles, and SWM planning. Since this study reviews older literature, there is hardly any study that has used CNN. This review emphasizes that AI based models face challenge of data shortage and less interpretability. [3] performed a systematic literature review in Solid Waste Management (SWM) on 40 studies published from 2019 to 2022. They compared these studies for the DL models applied on various datasets. A review specifically focused on application of object detection, Deep Learning (DL) models and allied datasets is done by [4]. They have exhaustively analyzed 10 waste detection, 15 CNN and YOLO based waste classification and 20 waste datasets. These datasets cater to different tasks respect to SWM such as underwater waste, data collected by unmanned drone etc. Not all datasets are available freely. These review articles highlight unavailability of freely available and labelled benchmark datasets in SWM. As a result, research community is inclined towards collecting their own dataset. This hinders comparison among various studies.

Considering this research gap, [5] released TrashNet dataset that consists of 2527 images and 6 categories such as paper, glass, plastic, metal, cardboard and trash. Due to its free availability, this dataset is used in many studies. [6] used TrashNet [5] dataset and variety of CNN architectures such as Densenet121, DenseNet169, InceptionResnetV2, MobileNet and Xception with various epochs. The best results were obtained with DenseNet121 using fine-tuning with a test accuracy rate of 95% and with the InceptionResNetV2 model using fine-tuning with a test accuracy of 94%. The drawback of this study is that variety of datasets are not experimented with. [7] used ResNet50 [?] CNN model as a feature extractor and SVM as a classifier on TrashNet [5] dataset and have got 87% accuracy. They have not compared their results with any other studies. A comparative study of CNN based models for waste classification is done by [8]. They have used ResNet18 [9], ResNet50 [10] and VGG16 [11] on TrashNet dataset. Since training deep models gets computationally expensive, they

finetuned the pretrained models. The best accuracy of 87% was obtained with ResNet18 model.

With advent of IOT and cloud technologies, several attempts are made to design smart bins and smart waste management systems. These systems make use of CNN models for task of waste classification. To tackle problem of waste segregation from its source, [12] have proposed as smart bin which uses camera, ultrasonic, infrared sensor and capacitive plates to detect if someone is trying to throw waste and to segregate it as dry and waste. A Raspberry pi module achieves the automatic operation of bin lid and flap that achieves segregation. Though this is innovative solution, it cannot handle mixed waste. There is no mention of object detection framework and CNN architecture used. Also, no thorough experimentation is done of this system. An IoT based smart bin prototype is developed by [13] in which an ultrasonic, GPS, RFID and camera sensors are used to detect and segregate waste. A lightweight MobileNetV2 [14] CNN model is used for waste classification which is incorporated in Raspberry Pi 3 module. Object detection for the waste is done using pre-trained Single-shot multi-box detector (SSD) MobileNetV2 TensorFlow object detection API running on the Raspberry Pi. It is fine-tuned using 365 images of waste which are collected by authors and labelled using Labellmg API. Lightweight long range LoRa protocol is used to communicate collected data. This prototype is compact but it needs to get trained on variety of data.

A comparison between deep and non-deep classification model is done by [10]. They have used Decision Tree, Random Forest, SVM and ResNet34 [9] models with fine tuning on TrashNet [5] dataset. The maximum accuracy obtained was 90% with CNN based ResNet34 model. Another smart bin is designed by [15] with the integration of the CNN model, SSD MobileNetV2 Quantized  $300 \times 300$  and Pi Camera on Raspberry Pi. The bin classifies 5 types of wastes, including metal, glass, paper, cardboard and plastic, with acceptable precision. Servo motors connected to the plastic board in the smart bin categorize the waste from temporary waste placement into the respective waste compartment. This process takes 4 seconds. SSD MobileNet2 object detection model is available in TensorFlow and is trained on MSCOCO dataset [16]. This classifies images at segment level. A smart municipal waste management system (MWM) is developed by [17] that uses sensor for waste level monitoring, CNN model deployed on cloud for waste classification. To reduce processing time, waste classification can also be done at the waste bin with reduced accuracy. The MWM can be monitored with mobile phone app. In order to choose an appropriate CNN model, they compared MobileNetV2, MobileNetV3, InceptionV3, ResNet50, ResNet101, ResNet152 and Xception models. They augmented the TrashNet dataset using web-scrapped waste images. The best performance in terms of accuracy and execution time was observed by MobileNetV3 with accuracy of 94.12% and running time as 261.7 ms.

Since TrashNet dataset works at image level and images are taken in clear background, some other datasets have emerged which handle different categories or segment level classification. With increase in plastic waste around the globe, WaDaBa dataset is created by [18]. The authors used

this dataset for PET classification using CNN [19]. This dataset is created with the aim of classifying different types of plastics into PET categories. [20] have developed a dataset MJU-Waste for the waste segmentation task. They have proposed a multi-level approach to waste object localization. Specifically, the method integrates the appearance and the depth information from three levels of spatial granularity. TACO dataset is developed in [21] which also deals with image segmentation. Both MJU-Waste and TACO are freely available. While, a permission has to be sought for using WaDaBa dataset from the authors.

In other notable contributions, a novel attempt is made to reduce waste produced due to low marketability of the agricultural products, [22] proposed a Batch Normalized CNN architecture that is used for classification of carrot fruit images. They prepared their own dataset by capturing images of carrots and data augmentation. They analyzed three CNN architectures out of which Batch Normalized architecture achieved accuracy of 99.43% on images of size 24 x 24. A multi-path CNN approach is presented in [23] for determining illegal dump sites. Due to unavailability of dump site dataset, [23] have prepared their own dataset.

In the attempt of biodegradable and non-biodegradable waste classification, authors of [25] have performed a brief review of 20 research articles focusing on applications of machine learning (ML), deep learning (DL), and IoT sensors in the waste classification. This work demonstrates that these technologies have shown promising advances for waste sorting accuracy, reducing human labor and promoting recycling efforts. Authors of [26] have experimented with only VGG16 model. Their model experiences overfitting and they lack experimentation with other state-of-the-art models. A biodegradable and non-biodegradable classification network (BnBCNet) is proposed in [27] that uses gated recurrent units (GRU) to capture enhanced frequent patterns with local and global context information. They have employed grey wolf optimization for fine tuning. Though their results are promising, they lack discussion on computational complexity of the model.

Literature review shows that most of the above studies fall into image classification category and very few fall in segmentation or pixel level classification category. Some of the gaps which are highlighted are as follows: Most of the studies have used image level classification with TrashNet dataset. This dataset has images taken with clear background. However, getting such setting in real life is difficult. With advances in IOT and edge computing, it is the need of the hour to have light weight CNN models which can give moderate classification accuracy. Waste has many categories. There is need of CNN models and datasets which can handle maximum variety in the waste. Though work is done for biodegradable and non-biodegradable waste, studies focusing on accuracy lack discussion on model complexity. Moreover, no comparative study is available for segmentation classification task.

### III. PROPOSED METHODOLOGY

The proposed methodology is the standard methodology followed in machine learning.

#### A. Dataset

Two primary sources are used to develop dataset:

##### 1) Custom dataset

Over 300 high-resolution images of  $3072 \times 4096$  pixels were captured using mobile phones to develop a custom dataset, supplemented with publicly available images sourced from Kaggle. Mobile phone configuration used is (i) Oppo Reno12 pro 5g with 42mm focal length, ISO 80, f/1.8 aperture and (ii) Samsung SM-A536E with 24mm focal length, ISO 400, f/1.8 aperture. Two separate phones were selected to bring the variety in resolution and picture quality. The dataset includes a wide range of biodegradable materials—such as food scraps, plant matter, and paper-based packaging—as well as non-biodegradable contaminants like plastic wrappers and synthetic materials. Images were collected under varied real-world conditions, including different lighting environments, angles, backgrounds, and stages of waste degradation, to introduce realistic complexity and simulate practical deployment scenarios. Annotation and class labeling were performed using the Roboflow platform<sup>2</sup>. The custom dataset is available at <https://universe.roboflow.com/sustainsort-dynamics/waste-detection-classification-eyd3h>. Some of the images from this dataset for class biodegradable are show in Fig. 1.

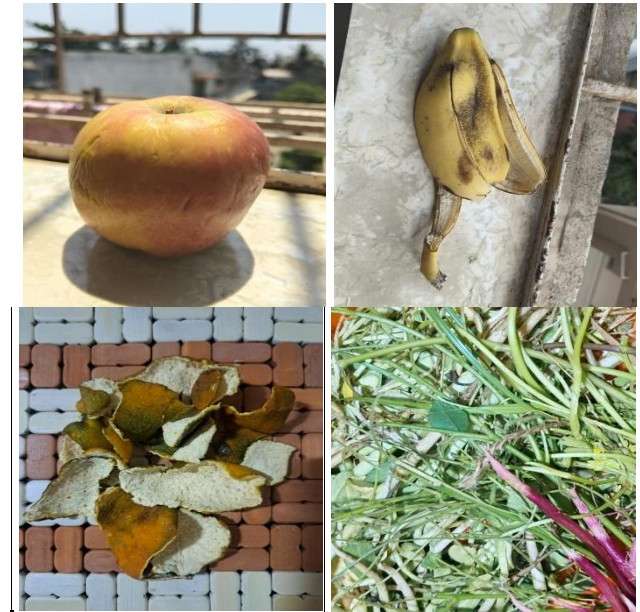


Fig. 1. Sample images for biodegradable class from custom dataset.

##### 2) Publicly available dataset

In order to bring in diversity in the custom dataset, 50% of image were taken from publicly available datasets on Kaggle. These available images enhanced augmentation and diversity of dataset.

##### 3) Preprocessing

<sup>2</sup> <https://roboflow.com/>

Preprocessing was performed using for preparing computer vision datasets. The preprocessing pipeline included: Normalization, resizing and noise reduction. Normalization transformed pixel values to a standard scale of (0–1). Resizing of images was performed to fit the input dimension requirements of the selected CNN models (e.g.,  $224 \times 224$  pixels). Noise Reduction is performed to minimize background clutter and enhance feature clarity. Variability in the images was introduced at the time of taking the images with varied camera setting. This step produced clean, standardized, and enriched image data for training.

### B. Baselines

Since a customized data was collected in this work, state-of-the-art models from the literature were selected to validate their performance on this dataset.

#### 1) VGG16 and VGG19 [11]:

These models are developed by Visual Geometry Group and they are pre-trained on ImageNet 2014 dataset. Very small  $3 \times 3$  convolutional filters are used to push depth of CNN architectures to 16 and 19 layers in VGG16 and VGG19 respectively. These models are used for waste classification in [8].

#### 2) DenseNet121 [24]:

DenseNet architectures connect each layer to every other layer in a feed-forward fashion. This facilitates efficient training of the CNN model. This model is used in the literature [5]. Since this is a very deep model, it was fine-tuned with the data. Thus, number of trainable parameters are less in table I.

#### 3) ResNet18 [9]:

ResNet is neural network that eliminate vanishing gradient problem in deep networks using "skip connection". It enhances performance in task like image recognition as it allows training much deeper network.

#### 4) MobileNetV2 [14]:

MobileNetV2 is proposed for resource constrained devices. It improves the state of the art for visual recognition including classification and object detection using such devices. MobileNetV2 uses depth-wise separable convolution as efficient building blocks.

TABLE I. MODELS USED FOR EXPERIMENTATION

| Model Name  | # Parameters | # Trainable parameters |
|-------------|--------------|------------------------|
| VGG16       | 134.27M      | 134.270M               |
| VGG19       | 139.58M      | 119.55M                |
| DenseNet121 | 6.96M        | 0.002M                 |
| ResNet18    | 11.18M       | 11.180M                |
| MobileNetV2 | 2.23M        | 2.230M                 |

### C. Evaluation Metric

Performance of the models is validated using precision, recall, F<sub>1</sub>-measure and ROC-AUC curve.

## IV. EXPERIMENTATION AND RESULTS

The dataset has 905 images with 451 images of biodegradable class and 454 of non-biodegradable class. This ratio ensures balanced data. Dataset is split as train-test and

validation in 80%-10% and 10% respectively. Experimental set-up involves Google Colab CUDA Device Tesla T4 GPU<sup>3</sup>. State-of-the-art models mentioned in section III.B are fine-tuned and their performance is analyzed on the mentioned platform.

### A. Hyperparameter tuning

All models mentioned in III.B underwent hyperparameter tuning. Table II shows hyperparameter values and validation accuracy achieved with these parameters. It is observed that validation accuracy is higher for VGG models. ResNet18 model and MonileNetV2 can be trained with slightly higher learning rate and batch size.

TABLE II. HYPERPARAMETER VALUES

| Model       | Learning Rate | Optimizer | Batch Size |
|-------------|---------------|-----------|------------|
| VGG16       | 0.0001        | Adam      | 16         |
| VGG19       | 0.0001        | Adam      | 32         |
| DenseNet121 | 0.0001        | Adam      | 32         |
| ResNet18    | 0.0005        | Adam      | 64         |
| MobileNetV2 | 0.0005        | Adam      | 64         |

### B. Comparisson of computational complexity

Models under consideration were validated for training time and computational complexity. Giga Multiply-Accumulate Operations per Second (GMAC) is a metric to compute computation complexity of a machine learning model. Higher the GMACs, higher is the computation complexity. Table III denotes computational complexity in terms of GMAC and training time in minutes. It is observed that VGG models have higher computational complexity and training time. MobileNetV2 is the most suitable model for real-world deployment on resource constrained devices.

TABLE III. ANALYSIS OF COMPUTATIONAL COMPLEXITY AND TARINING TIME

| Model       | #Parameters (M) | GMAC  | Training time (min) |
|-------------|-----------------|-------|---------------------|
| VGG16       | 134.27          | 15.52 | 8.48                |
| VGG19       | 139.58          | 19.68 | 0.60                |
| DenseNet121 | 6.96            | 2.90  | 0.52                |
| ResNet18    | 11.18           | 1.82  | 6.32                |
| MobileNetV2 | 2.23            | 0.319 | 6.10                |



Fig. 2. Training time (min) v Number of Parameters (M) for all models

<sup>3</sup> [https://colab.research.google.com/github/d2l-ai/d2l-tvm-colab/blob/master/chapter\\_gpu\\_schedules/arch.ipynb](https://colab.research.google.com/github/d2l-ai/d2l-tvm-colab/blob/master/chapter_gpu_schedules/arch.ipynb)

### C. Analysis of accuracy

Table IV and V show the analysis of the models in terms of various evaluation metrics. MobileNetV2 model shows remarkably high precision, recall,  $F_1$ -score and AUC-ROC. VGG16 model is performing excellently overall, but it's slightly better at recalling training examples than generalizing to new ones — a sign of mild overfitting or natural data variability.

TABLE IV. ANALYSIS OF PRECISION, RECALL AND  $F_1$ -SCORE

| Model       | Precision | Recall | $F_1$ -score |
|-------------|-----------|--------|--------------|
| VGG16       | 0.96      | 0.96   | 0.96         |
| VGG19       | 0.91      | 0.90   | 0.90         |
| DenseNet121 | 0.96      | 0.96   | 0.96         |
| ResNet18    | 0.96      | 0.96   | 0.96         |
| MobileNetV2 | 0.98      | 0.98   | 0.98         |

TABLE V. ANALYSIS OF ACCURACY ON TRAINING, VALIDATION AND TEST DATA

| Model       | Train Accuracy | Test Accuracy | Validation Accuracy |
|-------------|----------------|---------------|---------------------|
| VGG16       | 100.00%        | 95.56%        | 98.90%              |
| VGG19       | 99.03%         | 90.00%        | 97.96%              |
| DenseNet121 | 97.65%         | 95.56%        | 95.60%              |
| ResNet18    | 95.86%         | 95.56%        | 96.70%              |
| MobileNetV2 | 98.90%         | 97.78%        | 96.70%              |

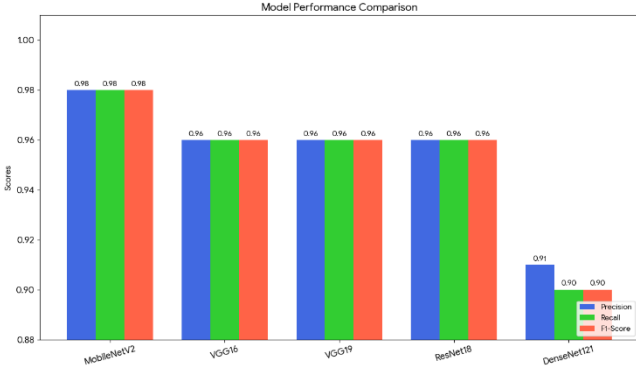


Fig. 3. Comparison based on evaluation metrics (Precision, Recall,  $F_1$ -score (%))

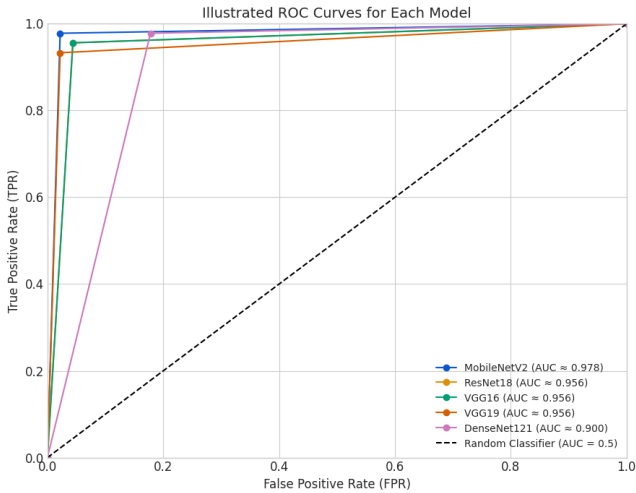


Fig. 4. ROC curve for all models (TPR vs FPR)

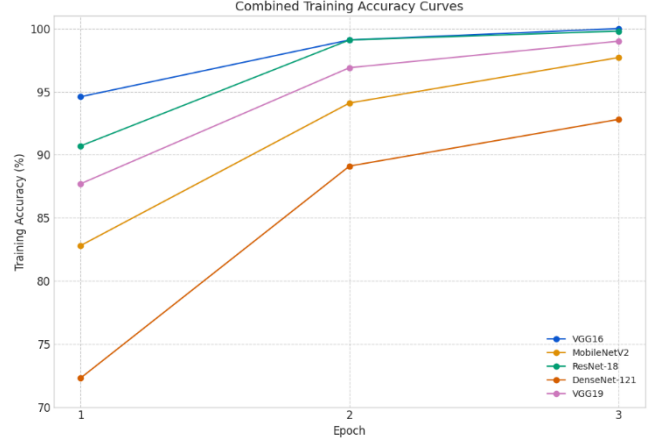


Fig. 5. Combined Training Accuracy Curves (Accuracy(%) vs Number of epochs)

### D. Robustness analysis

The models are evaluated for their robustness by adding random noise and brightness conditions for simulating real-world imperfections. Almost all models show slight drop in the accuracy. Refer Fig 6 and 7. Maximum drop is of 2.5% for VGG models while other models have negligible drop. This shows that models are already robust for noisy data. This could be because of two possible reasons – (1) Images were captured by two separate cameras which captured varied resolution and brightness conditions. (2) The data augmentation performed at the time of dataset creation added variability in the data which helped in achieving model robustness.

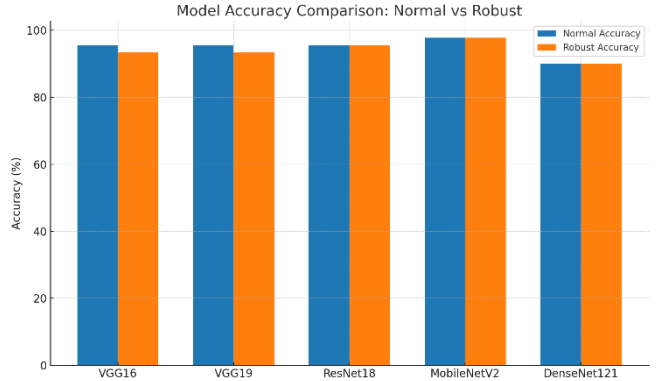


Fig. 6. Model Accuracy Comparison : Normal vs Robust

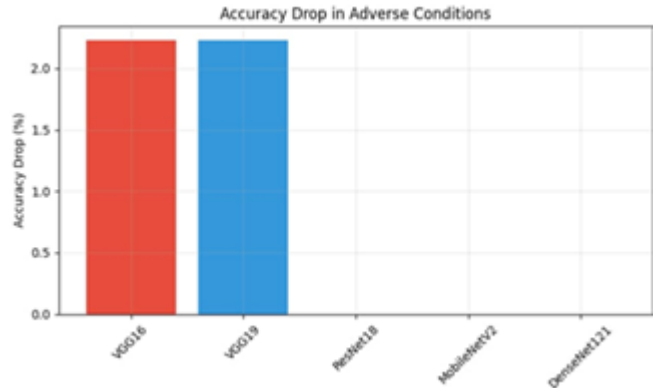


Fig. 7. Drop in model accuracy in adverse conditions



## V. CONCLUSION AND FUTURE SCOPE

In this work, we have attempted to classify waste based on images in biodegradable and non-biodegradable classes. We gathered data of household waste and augmented it with publicly available datasets. We then trained state of the art deep CNN models on this data. We evaluated performance of these models with respect to model size in terms of number of parameters, accuracy, computation efficiency and robustness. It is observed that VGG models viz. VGG16 and VGG19 have high accuracy but they are not computationally efficient. They are also susceptible to deviation in the performance in presence of noisy data. MobilenteV2 has reasonably high test accuracy and at the same time it is also computationally efficient. It's performance also does not drop in presence of noise. This shows that if the task of image classification is to be performed in a resource constrained environment, MobileNetV2 model is a preferred choice and it is suitable for deployment. DenseNet121 and ResNet18 show moderate performance on all frontiers.

One of the limitations of this work is limited data. Going ahead, we aim at increasing the dataset size to capture multiple types of biodegradable wastes. Alternatively, the dataset size can be increased by data augmentation techniques like flip, rotation, scaling etc. One of the future directions of this work is multi-class classification depending on waste categories. We also aim at integrating this work with robotics and related hardware to build a real time waste segregation system that can be utilized by municipal authorities. Another future direction of this work is to consider multi-modal learning.

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